

Madhava Lakshmi Satya Mohan Samanthakurthi

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PROFESSIONAL SUMMARY

Certified **Azure Data Engineer** and **Power BI Data Analyst** with 3+ years of enterprise experience at Cognizant, delivering end-to-end data engineering and BI solutions across finance and healthcare domains. Strong expertise in on-prem SQL Server, SSIS, SSRS, and large scale migrations to Azure cloud services, Databricks, PySpark, Azure Data Factory, Microsoft Fabric, lakehouse, scalable data pipelines, and analytics-ready data models and Power BI. Proven Technical POC and Lead Developer, driving modern data platform validation, data quality checks, KPI reporting, and stakeholder-ready dashboards. Recently completed an MSc in Artificial Intelligence for Business, applying analytics and AI-driven healthcare systems solutions (**Stamp 1G Visa**).

PROFESSIONAL EXPERIENCE

Cognizant, Associate Engineer, Hyderabad, India

Jul 2024 – Dec 2024

Client: Optum Finance (United Health Group)

- Engineered Azure Databricks workflows and Power BI dashboards using PySpark, Azure SQL, and DAX, enabling finance stakeholders to monitor **KPIs in real time** and reduce manual Excel-based reporting by nearly 30%.
- Developed interactive Power BI dashboards using DAX measures, Power Query transformations, and Row-Level Security (RLS), translating reporting requirements into stakeholder-ready visuals and drill-down insights.
- Designed and optimized large-scale PySpark transformation pipelines (joins, aggregations, standardization) across Medalion architecture layers, ensuring analytics-ready datasets for downstream reporting.
- Performed SQL based data analysis and reconciliation checks to validate business rules, identify reporting inconsistencies, and improve data accuracy for executive and operational reporting.
- Migrated 50+ SQL scripts to Azure Databricks Notebooks, automating batch data loads via workflows and configuration tables, improving execution **performance by 25%** and reducing operational dependency.
- Automated 20+ Jaspersoft and SSRS reports** with scheduled executions, ensuring 100% on-time financial reporting and improving reporting reliability under strict client SLAs.

Cognizant, Associate Engineer, Hyderabad, India

Oct 2023 – Jun 2024

Client: CareSource – BI Reporting 2.0

- Led Proof of Concept (POC) and served as Technical POC for migration from on-prem SSIS and SSRS to Azure Databricks SQL within the **Modern Data Platform (MDP)**, validating cloud feasibility and accelerating enterprise cloud adoption.
- Served as Data Analyst and Technical POC, performing comparative data analysis and KPI validation between legacy and modern platforms to ensure consistent stakeholder reporting.
- Performed comparative data analysis and KPI validation between legacy SQL Server and Azure Databricks environments, reconciling 10M+ records and **achieving 98.5%** data accuracy before go-live.
- Designed cloud-native data architecture within 1 month of Azure onboarding, replicating legacy SSIS packages and SSRS reports into Databricks and Power BI, improving **scalability and system performance**.
- Designed Power BI dashboards and SSRS reports to provide visibility into claims and finance operations, enabling business users to track trends across multiple U.S. states.
- Re-engineered SSIS ETL workflows into Databricks SQL and PySpark based ELT pipelines, orchestrated using **Databricks workflows**.

Cognizant, Programmer Analyst, Hyderabad, India

Nov 2021 – Sep 2023

Client: CareSource – Reporting

- Developed and optimized complex T-SQL stored procedures and SSIS ETL pipelines processing high-volume finance and claims datasets, improving query **performance by 20%** through indexing and tuning strategies.
- Developed complex T-SQL queries, stored procedures, and views to support ad-hoc analysis, stakeholder reporting, and recurring reporting needs across finance and claims domains.
- Designed and deployed 30+ parameterized SSRS reports** (drill-down, matrix, dynamic formatting), enhancing business visibility across multi-state healthcare operations.
- Built robust ETL solutions integrating heterogeneous data sources (SQL Server, flat files, Excel, ODBC/OLE DB), ensuring data consistency and integrity across reporting systems.
- Automated bulk data operations using advanced SSIS control flows and transformations (For Loop, For Each, Execute SQL, Bulk Insert) to automate complex finance and claims data workflows.
- Managed deployments using Git and CI/CD pipelines, ensuring stable releases and **seamless operations across** Indiana, Ohio, West Virginia, and Georgia business domains.
- Designed indexed views, triggers, user-defined functions, and constraints in SQL Server to enforce data integrity and improve transactional performance.

EDUCATION AND ACADEMIC PROJECTS

MSc in AI for Business, National College of Ireland, Dublin

Jan 2025 – Jan 2026

- Automated the **Leap Card student application process** using UiPath (RPA) with rule-based validation, CSV audit reports, email notifications.
- completed MSc research on Hybrid Ant Colony Optimization For Last Mile Delivery using ANN, KNN, and LSTM with Python and ML frameworks for last-mile delivery optimization.

Bachelor of Technology, Electronics and Communication Engineering, SRIN College, India

June 2016 – Nov 2020

SKILLS & TECHNOLOGIES

Cloud Skills

Azure Data Factory, Azure Databricks, Microsoft Fabric, Lakehouse, OneLake, Azure SQL, Power BI Service and Desktop, Data modeling, Event Streams, Data Activator, KQL Database, Lakehouse Medallion architecture.

Business Intelligence | Data Engineering

T-SQL, MS SQL Server, SSMS, SSIS, SSRS Paginated Reports, Power BI (DAX, Power Query), Tidal Automation, Jaspersoft, Rapid-Miner, ADF Pipelines, Databricks Workflows, ETL/ELT pipelines and batch processing.

Programming | ML

SQL, PySpark, Python, KQL (Kusto Query Language), NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Supervised, Unsupervised, Reinforcement Learning, Recommendation Systems, Feature Engineering, Model Evaluation, Feature Scaling, NLP, ANN, DNN.

Business Tools | DevOps

Jupyter Notebook, Google Colab, VS Code, Git, Databricks Workflows, ADF Pipelines, Tidal Automation, Jira, Confluence, Agile delivery (Scrum, Kanban), Data governance and risk compliance.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft | Dec 2024
- Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft | Dec 2024
- Microsoft Certified: Power BI Data Analyst (PL-300) | Microsoft | Dec 2024
- Microsoft Fabric Data Engineer Associate (DP-700) | Microsoft | March 2026
- Supervised Machine Learning: Regression and Classification | DeepLearning.AI | Dec 2025
- Advanced Learning Algorithms | DeepLearning.AI | Jan 2026
- Unsupervised Learning, Recommenders, Reinforcement Learning | DeepLearning.AI | Jan 2026
- Oracle Cloud Infrastructure AI Foundations Associate (1Z0-1122-25) | Oracle | Sep 2025
- Programming for Everybody (Getting Started with Python) | University of Michigan (Coursera) | Aug 2025.
- 20+ Professional Courses in SQL, SSIS, ADF, Databricks, Power BI, PySpark, Fabric, kafka | Udemy · DataCamp · Coursera · Microsoft Learn | Ongoing

AWARDS & RECOGNITION

- Best Employee of the Project - Caresource MDP migration (2024).
- Emerging Talent Award (2023).
- “Working as One” Award (2022).